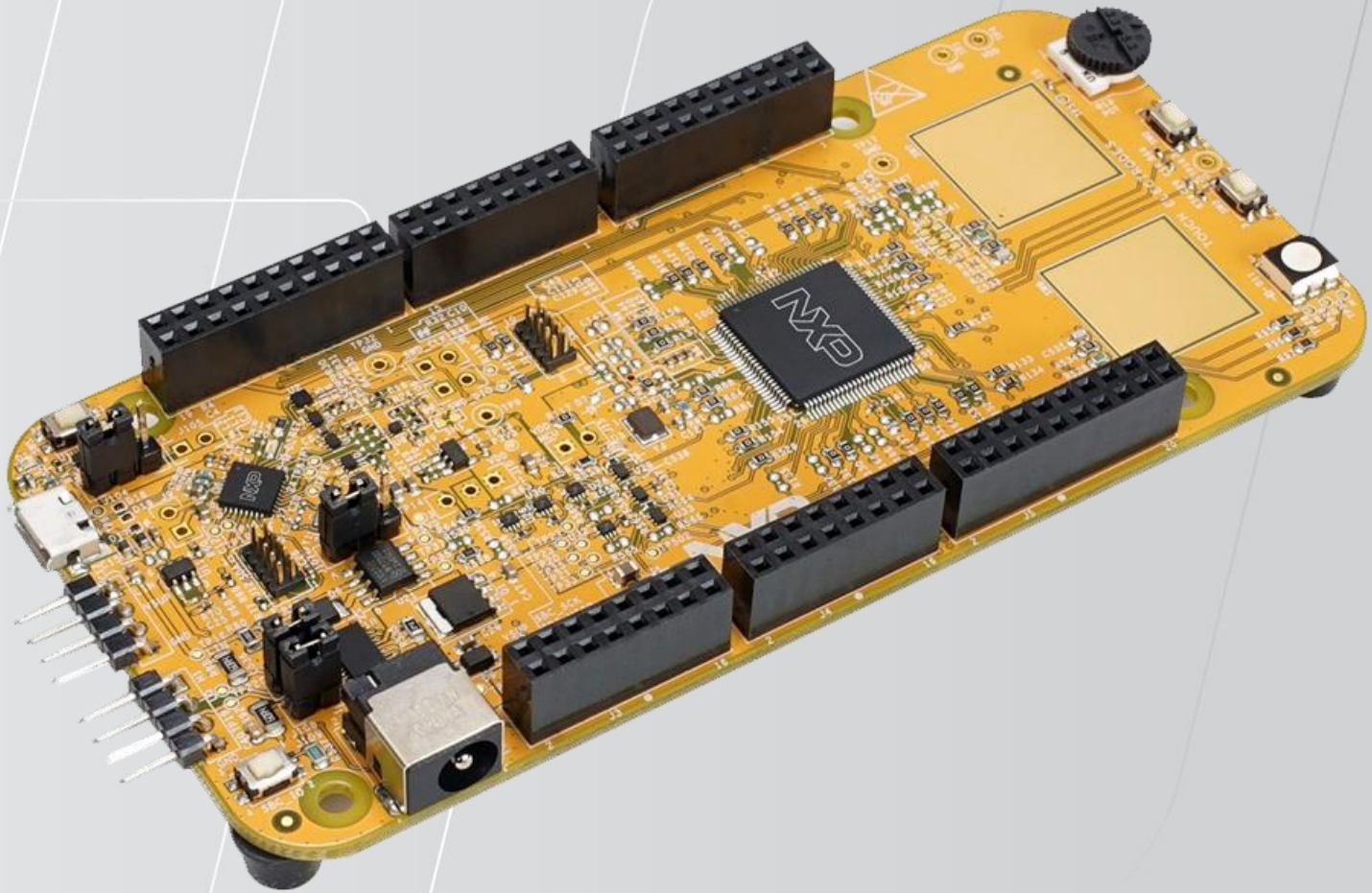
Timer



29 Nov 2022

OUTLINE

1. Timer Modules Overview
2. Low Power Interrupt Timer (LPIT)

1. Timer Modules Overview

The S32K144 supports following timer modules:

+ FlexTimer Module (FTM)
+ Low Power Interrupt Timer (LPIT)
+ Low Power Timer (LPTMR)
+ Real Time Clock (RTC)

2. Low Power Interrupt Timer (LPIT)

LPIT is a **low power periodic interrupt timer** with multiple timer channels. After a timer reaches a programmed count, the respective timer channel will generate pre-trigger and trigger output signals, and these pre-trigger and trigger outputs can be used to trigger other modules on the device.

Typical applications:

- To implement a clock
- To check user input periodically
- To monitor environment changes
- To switch between programs
- Count how many cycles/times the MCU has been elapsed since last restart.

2. Low Power Interrupt Timer (LPIT)

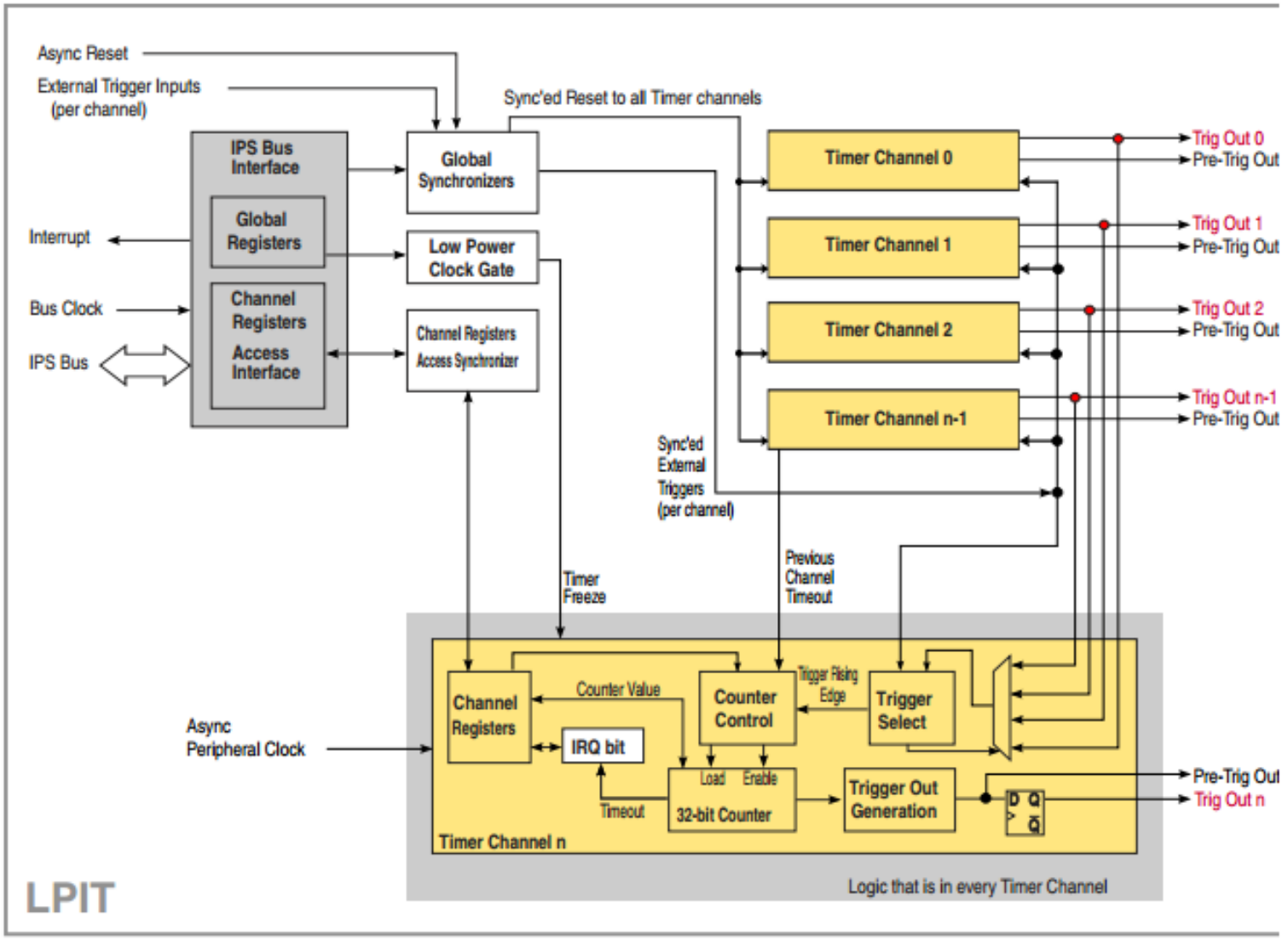


Figure 48-3. Detailed Block Diagram

2. Low Power Interrupt Timer (LPIT)

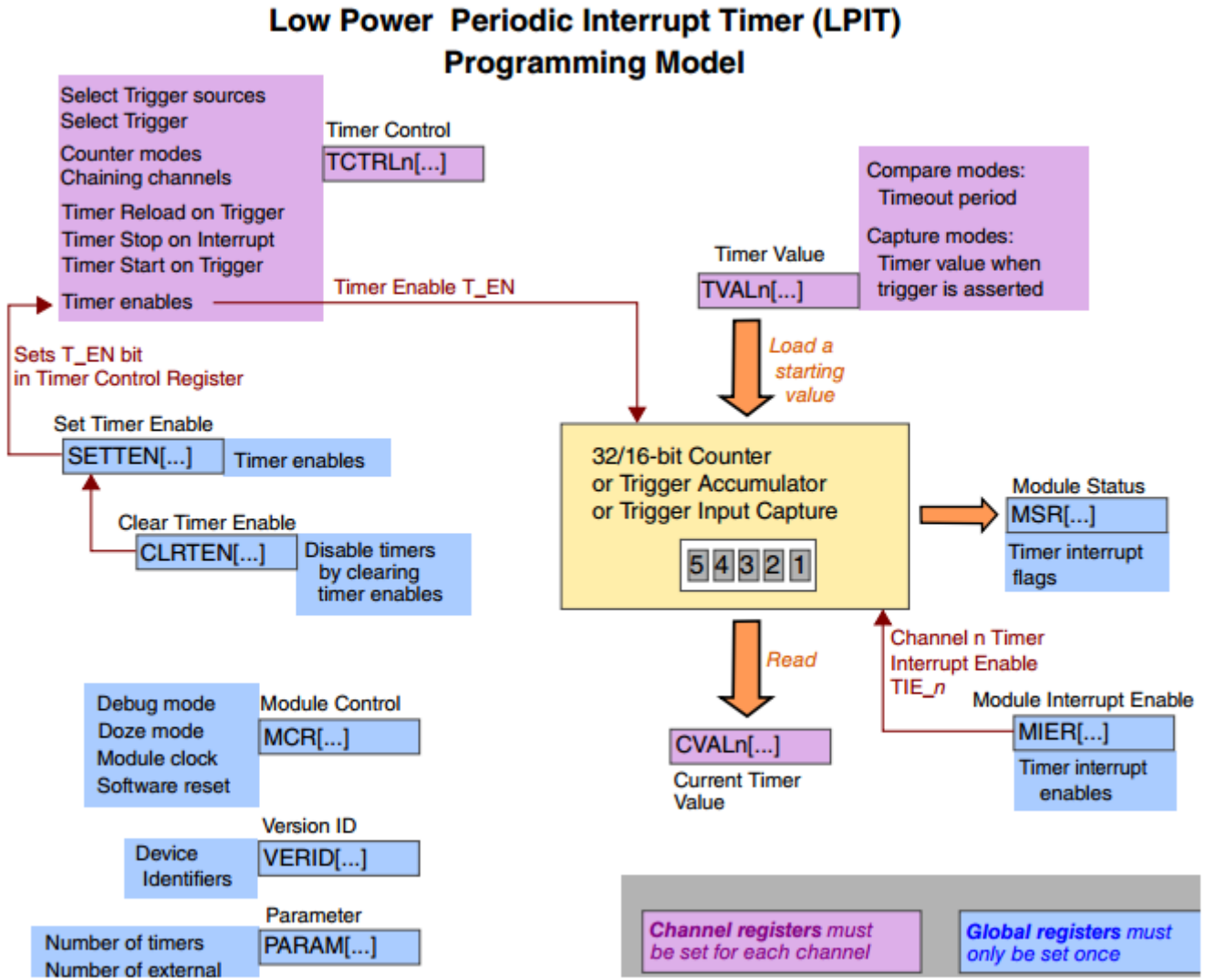
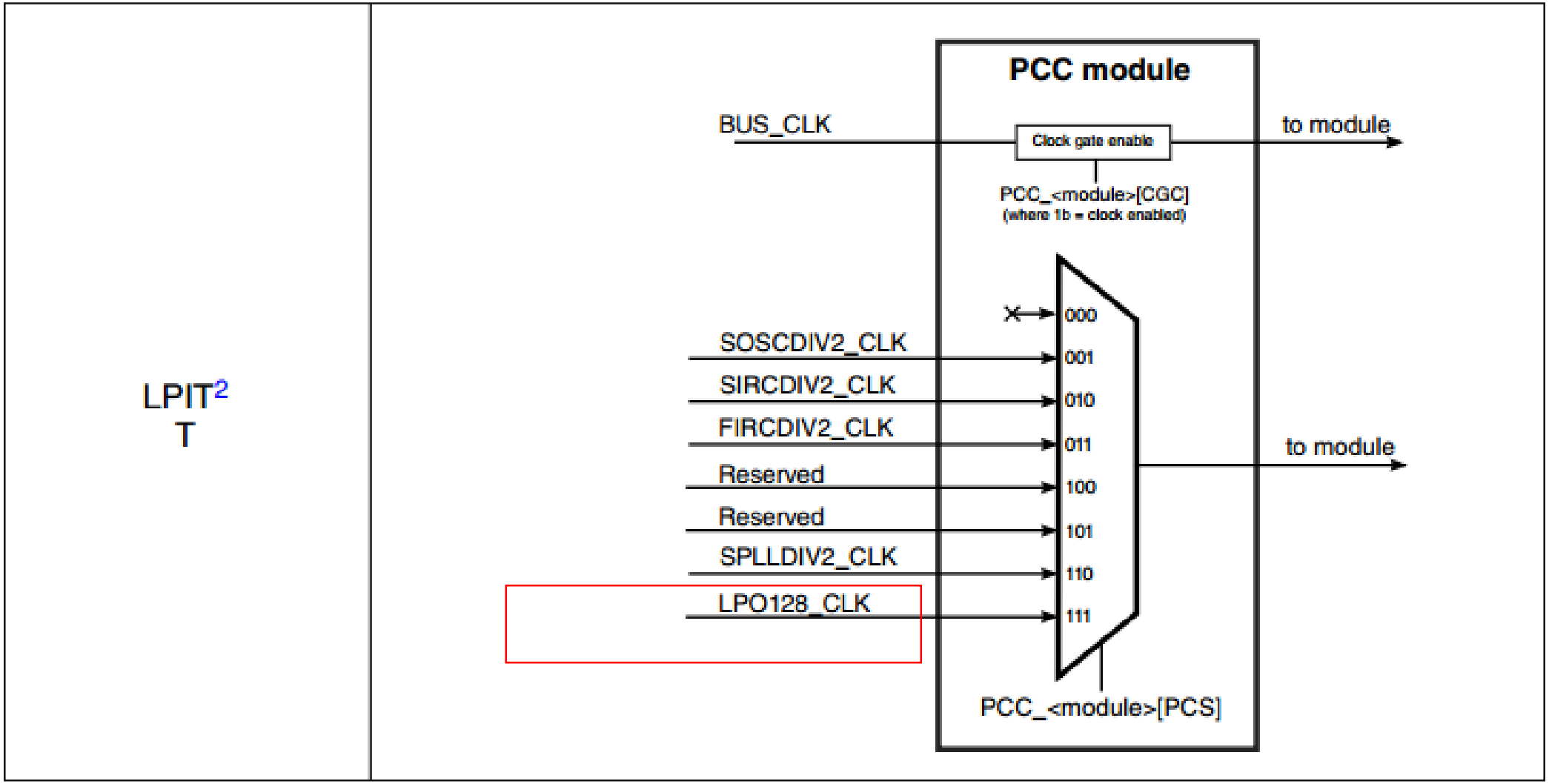


Figure 48-4. Programming Model

2.1. Source clock LPIT



2.2. Timer Modes LPIT

Timer Mode	Operation
32-bit Periodic Counter	• The counter will load, decrement down to 0 • which will set the timer interrupt flag and assert the output pre-trigger.
Dual 16-bit Periodic Counter	• The counter will load, then the lower 16-bits will decrement down to 0 • which will assert the output pre-trigger. • The upper 16-bits will then decrement down to 0 • which will set the timer interrupt flag and then negate the output pre-trigger.
32-bit Trigger Accumulator	• The counter will load on the 1st trigger rising edge and then decrement down to 0 on each trigger rising edge • which will set the timer interrupt flag and assert the output pre-trigger.
32-bit Trigger Input Capture	• The counter will load with 0xFFFF_FFFF and then decrement down to 0. • If a trigger rising edge is detected, ◦ then it will store the inverse of the current counter value in the timer value register ◦ which will set the timer interrupt flag and assert the output pre-trigger.

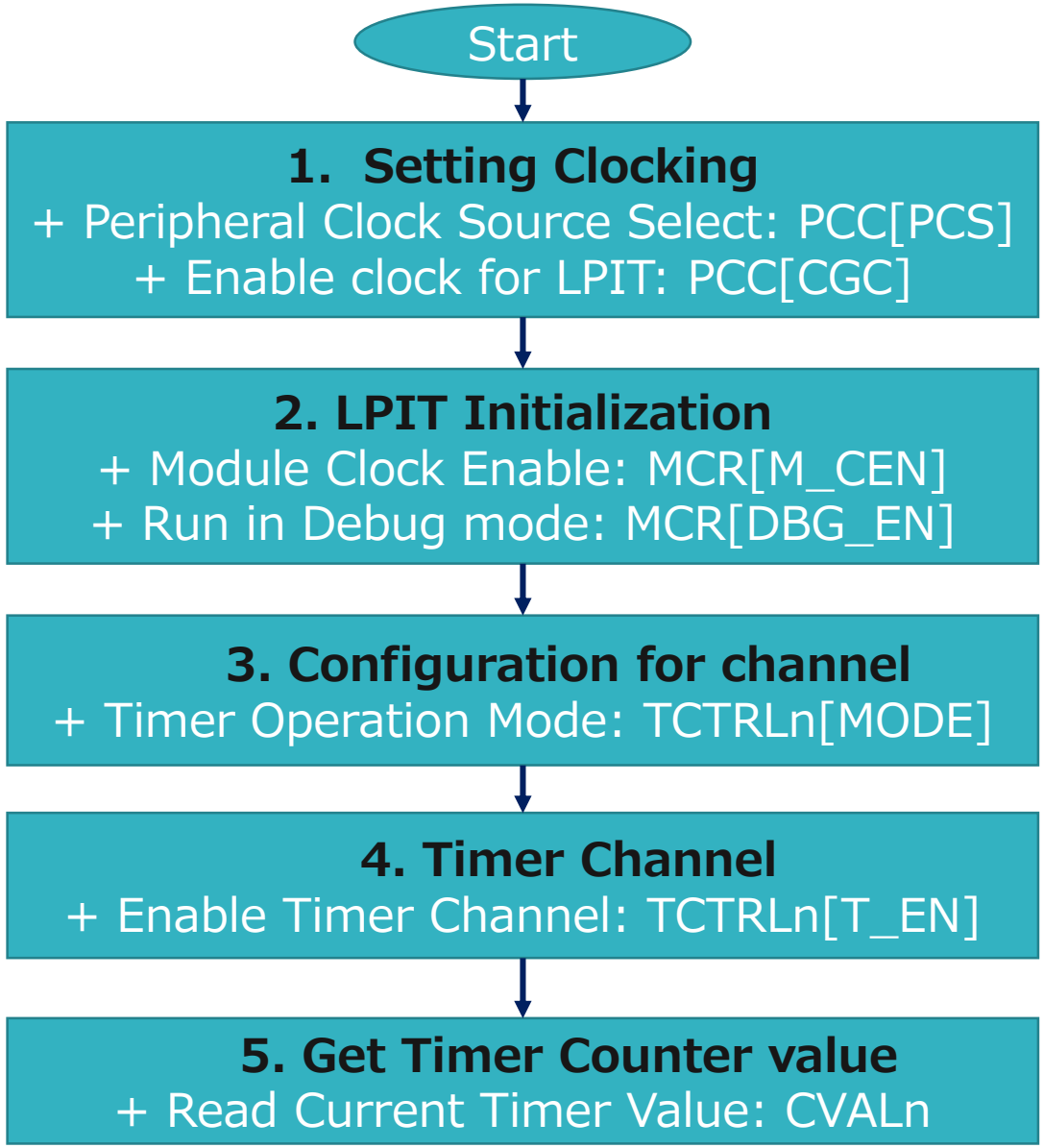
The timer mode is configured by setting an appropriate value in the MODE bits in TCTRLn register

2.3. LPIT memory map

LPIT base address: 4003_7000h

Offset	Register	Width (In bits)	Access	Reset value
0h	Version ID Register (VERID)	32	RO	0100_0000h
4h	Parameter Register (PARAM)	32	RO	0000_0404h
8h	Module Control Register (MCR)	32	RW	0000_0000h
Ch	Module Status Register (MSR)	32	W1C	0000_0000h
10h	Module Interrupt Enable Register (MIER)	32	RW	0000_0000h
14h	Set Timer Enable Register (SETTEN)	32	RW	0000_0000h
18h	Clear Timer Enable Register (CLRTEN)	32	WORZ	0000_0000h
20h	Timer Value Register (TVAL0)	32	RW	0000_0000h
24h	Current Timer Value (CVAL0)	32	RO	FFFF_FFFFh
28h	Timer Control Register (TCTRL0)	32	RW	0000_0000h
30h	Timer Value Register (TVAL1)	32	RW	0000_0000h
34h	Current Timer Value (CVAL1)	32	RO	FFFF_FFFFh
38h	Timer Control Register (TCTRL1)	32	RW	0000_0000h
40h	Timer Value Register (TVAL2)	32	RW	0000_0000h
44h	Current Timer Value (CVAL2)	32	RO	FFFF_FFFFh
48h	Timer Control Register (TCTRL2)	32	RW	0000_0000h
50h	Timer Value Register (TVAL3)	32	RW	0000_0000h
54h	Current Timer Value (CVAL3)	32	RO	FFFF_FFFFh
58h	Timer Control Register (TCTRL3)	32	RW	0000_0000h

2.4. Configuration for LPIT channel

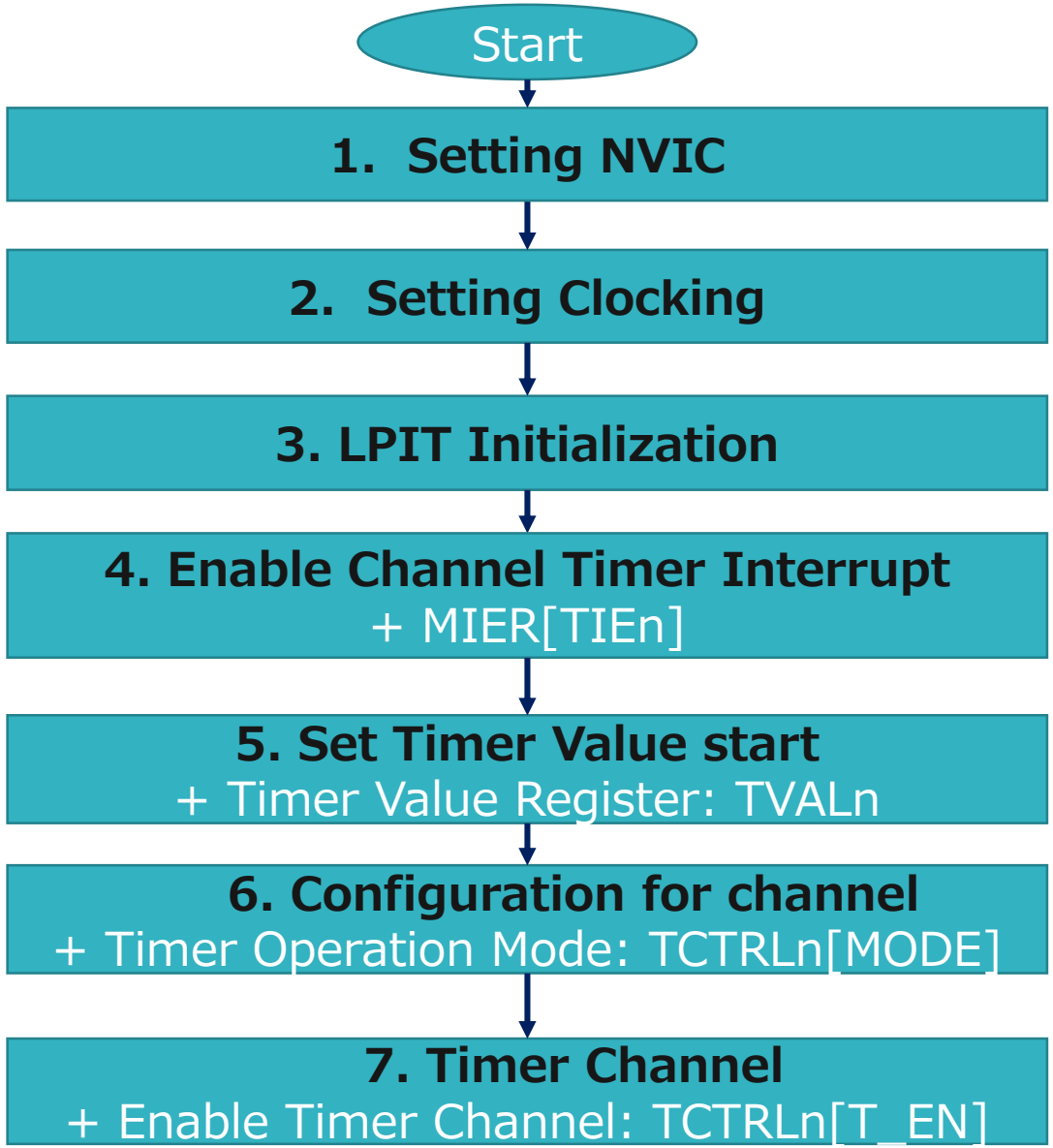


Measurement performance

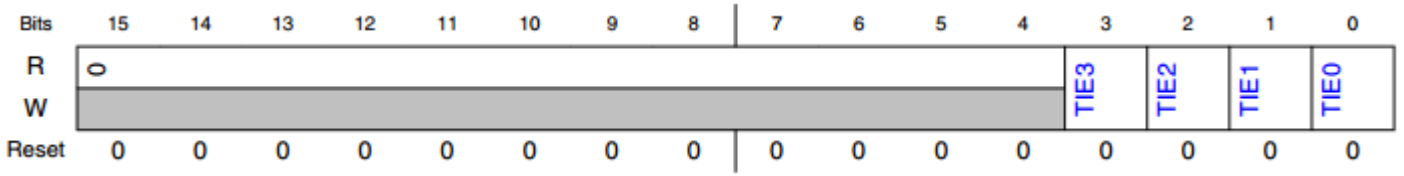
Measure the CPU time to complete this for loop.

```
for( int i = 0; i <100000; i++);
```

2.5. Configuration for timeout interrupt

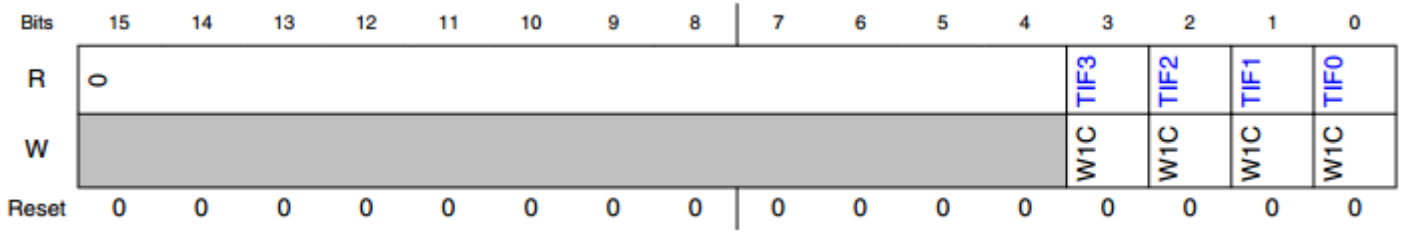


2.5.1. Module Interrupt Enable Register (MIER)



Field	Function
3 TIE3	Channel 3 Timer Interrupt Enable Enables interrupt generation when: • the Channel 3 Timer Interrupt Enable bit (TIE3) is set to 1 • and if the corresponding Timer Interrupt Flag (MSR[TIF3]) is asserted (=1, timeout has occurred) 0b - Disabled 1b - Enabled
2 TIE2	Channel 2 Timer Interrupt Enable Enables interrupt generation when: • the Channel 2 Timer Interrupt Enable bit (TIE2) is set to 1 • and if the corresponding Timer Interrupt Flag (MSR[TIF2]) is asserted (=1, timeout has occurred) 0b - Disabled 1b - Enabled
1 TIE1	Channel 1 Timer Interrupt Enable Enables interrupt generation when: • the Channel 1 Timer Interrupt Enable bit (TIE1) is set to 1 • and if the corresponding Timer Interrupt Flag (MSR[TIF1]) is asserted (=1, timeout has occurred) 0b - Disabled 1b - Enabled
0 TIE0	Channel 0 Timer Interrupt Enable Enables interrupt generation when: • the Channel 0 Timer Interrupt Enable bit (TIE0) is set to 1 • and if the corresponding Timer Interrupt Flag (MSR[TIF0]) is asserted (=1, timeout has occurred) 0b - Disabled 1b - Enabled

2.5.2. Module Status Register (MSR)



Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0												TIF3	TIF2	TIF1	TIF0
W													W1C	W1C	W1C	W1C
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Field	Function
3 TIF3	Channel 3 Timer Interrupt Flag • In compare modes: at the end of the timer period, the channel timer interrupt flag is set to 1 • In capture modes: when the trigger asserts, the channel timer interrupt flag is set to 1 • To clear a channel timer interrupt flag, write logic 1 to it • Writing 0 to a channel timer interrupt flag has no effect 0b - Timer has not timed out 1b - Timeout has occurred (timer has timed out)
2 TIF2	Channel 2 Timer Interrupt Flag • In compare modes: at the end of the timer period, the channel timer interrupt flag is set to 1 • In capture modes: when the trigger asserts, the channel timer interrupt flag is set to 1 • To clear a channel timer interrupt flag, write logic 1 to it • Writing 0 to a channel timer interrupt flag has no effect 0b - Timer has not timed out 1b - Timeout has occurred (timer has timed out)

2.6. Assignment 2

Timer channel 2 creates an interrupt every 500 ms.

2.7. Assignment 3

Timer channel 3 creates an interrupt every 1s.
→ Write a program simulate a watcher
hh:mm:ss


Thank you